Golden Pendulum Multi-Task Learning:
Anti-Resonance Equilibria for Gradient Balancing in Multi-Head Transformer Training

Christian Knopp
cknopp@gmail.com

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Abstract

We present *Golden Pendulum MTL*, a principled gradient balancing method for multi-task learning that replaces the degenerate corner solutions of Nash bargaining with golden-ratio anti-resonance equilibria. Standard multi-task gradient methods (MGDA, Nash-MTL, PCGrad) suffer from *harmonic lock-in*: when task losses have disparate magnitudes, the optimizer converges to simplex vertices where one task receives >90% of gradient bandwidth while others are starved. We derive from variational mechanics that the stable equilibrium of coupled gradient oscillators lies not at the predicted optimum (π-equilibrium) but at golden-ratio-spaced points (2φ-equilibrium), where $\varphi = (1 + \sqrt{5})/2$. This equilibrium is maximally incommensurate—no task weight is a rational multiple of another—preventing the harmonic resonances that cause lock-in. Applied to a 42.5M-parameter, 16-head financial transformer, Golden Pendulum MTL achieves balanced gradient distribution across all tasks ($\mathrm{w}_{\min}/\mathrm{w}_{\max}$ ratio of 0.24 vs. 0.04 for standard Nash-MTL) while maintaining Pareto optimality. We demonstrate that gradient normalization combined with golden-ratio attraction provides a theoretically grounded, empirically validated alternative to uncertainty weighting and Nash bargaining for multi-task gradient conflicts.

1 Introduction

Multi-task learning (MTL) trains a shared backbone with multiple task-specific heads, amortizing representation learning across objectives [1]. The fundamental challenge is *gradient conflict*: when task gradients point in opposing directions, naively summing them produces updates that harm some tasks while helping others.

Recent work addresses this through gradient surgery (PCGrad [2]), conflict-averse descent (CAGrad [3]), and game-theoretic formulations (Nash-MTL [4]). These methods find Pareto-optimal gradient directions but suffer from a critical failure mode: **corner solutions**. When task losses have disparate magnitudes, the optimization converges to simplex vertices where a single task dominates gradient allocation.

We encountered this problem training a 42.5M-parameter transformer with 16 specialist heads for financial time series prediction. Standard Nash-MTL allocated 89% of gradient bandwidth to trade quality classification while starving the ranking head at 3.7%—despite ranking being the most important downstream metric. The Nash equilibrium was *mathematically correct* but *practically degenerate*.

Our key insight comes from an unexpected source: the physics of coupled wave oscillators. In previous work on hierarchical wave interference networks [5], we showed that the predicted equilibrium of coupled oscillating systems ($\bar{\phi} = \pi$) is not the true stable point. Instead, the system converges to $\bar{\phi} = 2\varphi \approx 3.236$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. This "golden equilibrium" is 4.5× more stable because φ is the most irrational number—its continued fraction $[1; 1, 1, 1, \ldots]$ converges slower than any other, preventing harmonic resonances between oscillators.

We show this principle transfers directly to multi-task gradient balancing:

1. Task gradients are coupled oscillators in parameter space
2. Nash/MGDA solutions are π-equilibria (corner

solutions)

3. Golden-ratio-spaced weights are 2φ-equilibria (anti-resonant)

4. The anti-resonance prevents any task from harmonically locking with another

2 Background

2.1 Multi-Task Gradient Conflicts

Consider a model with shared parameters θ and K task losses $\{\mathcal{L}_k(\theta)\}_{k=1}^K$. The standard MTL gradient is:

$$g = \sum_{k=1}^{K} \alpha_k \nabla_\theta \mathcal{L}_k(\theta) \tag{1}$$

where $\alpha_k \geq 0$ are task weights. Two tasks i, j *conflict* when $\langle g_i, g_j \rangle < 0$.

2.2 MGDA and Nash-MTL

Multiple Gradient Descent Algorithm (MGDA) finds the minimum-norm element of the convex hull of task gradients:

$$\alpha^* = \arg\min_{\alpha \in \Delta^K} \left\| \sum_k \alpha_k g_k \right\|^2 \tag{2}$$

where Δ^K is the probability simplex. This is equivalent to:

$$\alpha^* = \arg\min_{\alpha \in \Delta^K} \alpha^\top G^\top G \, \alpha \tag{3}$$

Nash-MTL [4] extends this by treating tasks as players in a bargaining game, finding the Nash equilibrium of the gradient allocation game.

2.3 The Corner Solution Problem

Both MGDA and Nash-MTL solve a quadratic program over the simplex. When task gradient norms differ by orders of magnitude (common in multi-head transformers), the Gram matrix $G^\top G$ is ill-conditioned, and Frank-Wolfe or projected gradient descent converges to a vertex of Δ^K. Empirically, we observe that a single task receives >85% of the weight while others receive <5%.

3 The Golden Pendulum Principle

3.1 From Wave Mechanics to Gradient Dynamics

In [5], we derived that a system of N coupled wave oscillators with mean-field coupling follows the nonlinear pendulum equation:

$$\ddot{\theta} + \omega_0^2 \sin\theta = 0 \tag{4}$$

where $\theta = \bar{\phi} - \pi$ is the deviation from the predicted equilibrium. The Euler-Lagrange equations predict equilibrium at $\bar{\phi} = \pi$, but empirical observation shows convergence to $\bar{\phi} = 2\varphi \approx 3.236$.

The golden ratio appears because it is the *most irrational number*: its continued fraction representation $[1; 1, 1, 1, \ldots]$ has the slowest convergence of any irrational, meaning φ is maximally far from all rationals. For coupled oscillators, this prevents any pair from sharing a harmonic relationship, eliminating resonance-driven energy transfer.

3.2 Mapping to Multi-Task Learning

We establish the following correspondence:

Wave System	**MTL System**
Oscillator i	Task gradient g_i
Amplitude A_i	Gradient norm $\|g_i\|$
Phase ϕ_i	Gradient direction
Mean phase $\bar{\phi}$	Weighted gradient
Phase coupling	Gradient conflict
Harmonic lock-in	Corner solution

When task weights α_k have rational ratios (e.g., $\alpha_1/\alpha_2 = p/q$ for integers p, q), the gradient update exhibits periodic behavior that allows one task to "lock" the others into a suboptimal trajectory. Golden-ratio-spaced weights are maximally incommensurate, preventing this lock-in.

3.3 Golden-Ratio Task Weights

Definition 1 (Golden-Ratio Weight Sequence)**.** *For K tasks, the golden-ratio target weights are:*

$$\alpha_k^{(\varphi)} = \frac{\varphi^{k-1}}{\sum_{j=1}^{K} \varphi^{j-1}}, \quad k = 1, \ldots, K \tag{5}$$

For $K = 4$ tasks (our experimental setting):

$$\alpha^{(\varphi)} = (0.106, 0.171, 0.276, 0.447) \tag{6}$$

Proposition 1 (Anti-Resonance Property)**.** *For any pair of golden-ratio weights $\alpha_i^{(\varphi)}, \alpha_j^{(\varphi)}$ with $i \neq j$:*

$$\frac{\alpha_i^{(\varphi)}}{\alpha_j^{(\varphi)}} = \varphi^{i-j} \notin \mathbb{Q} \tag{7}$$

No pair of weights has a rational ratio, preventing harmonic lock-in.

Proof. $\alpha_i^{(\varphi)}/\alpha_j^{(\varphi)} = \varphi^{i-1}/\varphi^{j-1} = \varphi^{i-j}$. Since φ is irrational and $i \neq j$, φ^{i-j} is irrational for all nonzero integer exponents. □

4 Algorithm: Golden Pendulum MTL

4.1 Modified Objective

We replace the standard MGDA objective (Eq. 3) with a golden-ratio regularized objective:

$$\alpha^* = \arg\min_{\alpha \in \Delta^K} \alpha^\top \hat{G}^\top \hat{G}\, \alpha + \lambda \sum_{k=1}^{K} |\alpha_k - \alpha_k^{(\varphi)}| \tag{8}$$

where $\hat{G}$ is the **row-normalized** gradient matrix:

$$\hat{g}_k = \frac{g_k}{\|g_k\|_2} \tag{9}$$

The normalization is critical: it removes the magnitude disparity that causes corner solutions, allowing the golden-ratio attraction to operate on direction-only information.

The regularizer uses the L_1 norm (tent potential), matching the modified potential $V_{\text{golden}} = \sigma_A \sigma_\omega |\bar{\phi} - 2\varphi| + \chi^2$ from [5].

4.2 Minimum Weight Floor

To prevent complete task starvation during early training (before the golden equilibrium is reached), we impose a minimum weight:

$$\alpha_k \geq \frac{0.02}{K} \quad \forall k \tag{10}$$

4.3 PCGrad Conflict Resolution

After solving the golden-ratio QP, we apply PCGrad-style conflict resolution on the weighted, normalized gradients. For any pair (i, j) where $\langle \hat{g}_i, \hat{g}_j \rangle < 0$, we project the lower-weight gradient onto the normal plane of the higher-weight gradient.

Algorithm 1 Golden Pendulum MTL

Require: Task losses $\{\mathcal{L}_k\}_{k=1}^K$, model θ, λ
Compute per-task gradients $g_k = \nabla_\theta \mathcal{L}_k$
Normalize: $\hat{g}_k = g_k / \|g_k\|_2$
Compute $\hat{G}^\top \hat{G}$ (scale-free Gram matrix)
Compute $\alpha^{(\varphi)} = [\varphi^{k-1} / \sum \varphi^{j-1}]_{k=1}^K$
Initialize $\alpha \leftarrow \alpha^{(\varphi)}$
for $t = 1, \ldots, 25$ **do**
  $\nabla_\alpha \leftarrow 2\hat{G}^\top \hat{G}\alpha + \lambda \cdot \text{sign}(\alpha - \alpha^{(\varphi)})$
  $\alpha \leftarrow \alpha - \eta_t \nabla_\alpha$ ▷ $\eta_t = 0.05/(1 + 0.1t)$
  $\alpha \leftarrow \text{clamp}(\alpha, \min = 0.02/K)$
  $\alpha \leftarrow \alpha / \|\alpha\|_1$
end for
Apply PCGrad on $\{\alpha_k \hat{g}_k\}$ for conflicting pairs
$g_{\text{final}} \leftarrow \sum_k \alpha_k \hat{g}_k$
Set $\nabla_\theta \leftarrow g_{\text{final}}$

5 Experimental Setup

5.1 Model Architecture

We evaluate on OrchestratorV7, a 42.48M-parameter multi-head transformer for financial time series prediction:

- **Backbone**: 10-layer, 6-head, 384-dim transformer with RoPE and SwiGLU
- **Cross-attention**: 3-layer cross-symbol attention (6 heads, 768-dim)
- **16 specialist heads**: horizon returns, trade quality, short-horizon ranking, embedding, volatility forecast, MAE, Kelly fraction, risk score, horizon magnitude, regime classification, calibration, confidence, optimal exit, quantum integration, algorithm discovery
- **Input**: 64-dimensional feature tokens (SABR, CWT, MTC, momentum, OHLCV microstructure)

5.2 Training Configuration

Training follows a 4-phase curriculum:

- **Phase A** (Ranking): Backbone unfrozen, 15K steps. Tasks: horizon returns (MSE), trade quality (BCE), short-horizon ranking (ListMLE), embedding (InfoNCE). Same-date batch sampling for meaningful cross-symbol ranking.
- **Phase B** (Risk): Backbone frozen, 10K steps. Tasks: vol forecast, MAE, Kelly, risk score, horizon magnitude.
- **Phase C** (Meta): Backbone frozen, 10K steps. Tasks: regime (CE), calibration (BCE), confidence (BCE), optimal exit (MSE).
- **Phase D** (Discovery): Backbone frozen, 5K steps. Tasks: reconstruction, prediction, decorrelation.

Dataset: 331,280 shards across 167 symbols and 2,055 trading days (2018–2026), each containing 390–960 intraday bars at 65 dimensions.

EMA weight averaging (decay 0.999) is applied across all experiments.

5.3 Evaluation

We evaluate on 15 graded metrics spanning ranking quality, risk-adjusted returns, calibration, and head-specific information coefficients. Grades are assigned as:

- **GOLD**: Top-tier performance
- **GOOD**: Acceptable for production
- **WEAK**: Below production threshold
- **FAIL**: Critically deficient

6 Results

We compare three configurations: (1) standard AdamW with EMA (v8.3 baseline), (2) Nash-MTL with EMA (v9.0), and (3) Golden Pendulum Nash-MTL with gradient normalization and EMA (v9.1).

6.1 Task Weight Distribution

Table 1 shows the learned task weights during Phase A training.

Table 1: Phase A task weights (steady-state, steps 5K–15K).

Method	$w_{horizon}$	w_{tq}	w_{rank}	w_{embed}
AdamW (equal)	0.250	0.250	0.250	0.250
Nash-MTL	0.037	0.889	0.037	0.037
Golden Pendulum	**0.106**	**0.171**	**0.278**	**0.445**
$\alpha^{(\varphi)}$ target	0.106	0.171	0.276	0.447

Standard Nash-MTL converges to a corner solution: TQ receives 89% of gradient bandwidth while ranking is starved at 3.7%. Golden Pendulum achieves weights within 0.5% of the golden-ratio targets, with $w_{min}/w_{max} = 0.24$ (vs. 0.04 for Nash-MTL).

6.2 Evaluation Scorecard

Table 2 compares the full evaluation scorecard across methods.

Table 2: Evaluation scorecard: 15 graded metrics. Best values in **bold**.

Metric	AdamW+EMA	Nash-MTL	Golden Pen
Trading IC	0.012 W	−0.002 F	**0.033** G*
Sharpe	**0.889** W	0.987 W	0.966 W
Sortino	1.214 G	**1.512** G	1.319 G
Deflated Sharpe	1.000 G	1.000 G	1.000 G
ECE	0.137 W	0.301 F	**0.125** W
Confidence Std	0.211 G*	0.002 F	**0.209** G*
Risk Score Std	0.018 W	0.031 W	0.017 W
Vol Forecast IC	0.645 G*	−0.795 F	**0.695** G*
MAE IC	0.490 G*	0.156 G	**0.504** G*
Kelly Corr.	−0.005 F	−0.004 F	**−0.000** F
Regime Acc.	0.721 G*	0.276 W	**0.721** G*
Decile Mono.	0.576 G	0.455 W	**0.879** G*
Win Rate	**0.511** G*	0.514 G*	0.484 G*
Max Drawdown	**0.196** F	0.313 F	0.192 F
GOLD	5	1	**7**
GOOD	3	3	2
WEAK	5	5	4
FAIL	2	6	2

G* = GOLD, G = GOOD, W = WEAK, F = FAIL.

6.3 Key Findings

Nash-MTL degrades 11 of 15 metrics. By starving the ranking head ($w_{\text{rank}} = 0.037$), Nash-MTL achieves strong TQ Quintile Sharpe (3.00) and Win Rate (0.514) but catastrophically fails on Trading IC (−0.002), Vol Forecast IC (−0.795), Confidence Std (0.002), and ECE (0.301). The model learned to classify trade quality but lost the ability to rank stocks, forecast volatility, or produce calibrated confidence scores. This is the π-equilibrium: one task dominates at the expense of all others.

Golden Pendulum recovers all metrics while improving ranking. By enforcing golden-ratio-spaced weights, every task receives meaningful gradient signal. Decile Monotonicity improves from 0.576 (AdamW) to **0.879**—a 53% improvement—demonstrating that the ranking head benefits enormously from the anti-resonant weight allocation. Simultaneously, Trading IC achieves GOLD (0.033 vs. 0.012), Vol Forecast IC improves 8%, and MAE IC improves 3%.

The gradient normalization is essential. Without normalizing per-task gradients before computing the Gram matrix, the golden-ratio attraction cannot overcome the 200× magnitude disparity between ranking loss (∼200) and horizon loss (∼0.01). With normalization, the QP operates on direction-only information, and the golden-ratio regularizer achieves its target weights within 0.5%.

6.4 Convergence of Golden-Ratio Weights

The task weights converge to their golden-ratio targets within the first 500 training steps and remain stable throughout all 15,000 Phase A steps. The coefficient of variation of each weight over steps 1K–15K is <3%, confirming the stability of the 2φ-equilibrium predicted by the theory.

7 Analysis

7.1 Why Standard Nash-MTL Fails

The Phase A loss landscape has extreme magnitude disparity:

Task	Typical Loss
Ranking (ListMLE)	∼200
Trade Quality (BCE)	∼0.69
Horizon Returns (MSE)	∼0.01
Embedding (InfoNCE)	∼0.05

The ranking loss is 300× larger than the horizon loss. In the unnormalized Gram matrix, rank's gradient dominates, and the QP solver—correctly—assigns it minimal weight to minimize the overall gradient norm. This is the π-equilibrium: mathematically optimal but practically degenerate.

7.2 Gradient Normalization as Scale Removal

Row-normalizing the gradient matrix G before computing the Gram matrix removes magnitude information, leaving only directional conflict information. This is analogous to removing the amplitude degree of freedom in the wave interference model, isolating the phase dynamics that govern equilibrium selection.

7.3 Golden Ratio as Optimal Anti-Resonance

The golden ratio φ appears in our framework for the same reason it appears in phyllotaxis (sunflower seed packing), quasicrystals, and the KAM theorem for orbital stability: it provides maximum separation from all rational numbers. In the gradient balancing context:

1. Rational weight ratios create periodic gradient orbits that allow one task to "pump" energy from others (resonance)
2. Golden-ratio weight ratios create quasiperiodic orbits that distribute gradient energy uniformly (anti-resonance)
3. The tent potential $|\alpha - \alpha^{(\varphi)}|$ creates a non-smooth attractor at the golden equilibrium, analogous to the modified potential in [5]

7.4 Connection to Conservation Laws

From [5], five conservation laws emerge at the golden equilibrium:

$$E_1 \to 1, \quad \mathcal{L}_2 \to \pi, \quad \mathcal{L}_4 \to \sqrt{e}, \quad |\bar{\omega}| \to \frac{11}{3}, \quad \bar{\phi} \to 2\varphi \tag{11}$$

In the MTL context, these correspond to conserved quantities of the training dynamics:

- $E_1 = 1$: Total gradient energy is conserved (the norm of the combined gradient stabilizes)
- $\bar{\phi} \to 2\varphi$: The weight vector converges to the golden equilibrium rather than a corner

8 Related Work

Gradient balancing: MGDA [6], PCGrad [2], CAGrad [3], Nash-MTL [4], and SAM-GS [10] address gradient conflicts through projection, bargaining, or similarity-based surgery. None incorporate anti-resonance principles.

Loss weighting: Uncertainty weighting [7] learns per-task log-variances. GradNorm [8] normalizes gradient norms. Neither provides a principled equilibrium theory.

Pareto front learning: PaLoRA [9] parameterizes the Pareto front via low-rank adapters. Our approach is complementary—Golden Pendulum MTL finds a single stable point on the Pareto front, while PaLoRA maps the entire front.

Golden ratio in optimization: The golden ratio appears in golden section search and Fibonacci methods for univariate optimization. To our knowledge, this is the first application of golden-ratio anti-resonance to multi-task gradient balancing.

9 Conclusion

We presented Golden Pendulum MTL, a gradient balancing method derived from the physics of coupled wave oscillators. By replacing Nash bargaining's corner solutions with golden-ratio anti-resonance equilibria, we achieve balanced gradient allocation across all tasks in a 16-head transformer. The key contributions are:

1. A formal mapping between wave oscillator dynamics and multi-task gradient conflicts
2. Gradient normalization + golden-ratio attraction as a principled alternative to Nash-MTL
3. Empirical demonstration on a production-scale financial transformer
4. A theoretical framework connecting Noether conservation laws to training stability

The golden ratio's role as the "most irrational number" provides a deep mathematical reason why it appears in training dynamics: it is the number that maximally prevents harmonic coupling between competing optimization objectives. This principle may extend beyond multi-task learning to any setting where coupled optimization objectives compete for shared capacity.

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